

PINN optimisation for damped structural vibration: findings and path to railway predictive maintenance

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OBJECTIVE

One objective, three papers: understand PINN failure on a controlled beam, then transfer to railways

Paper	Scope	Status
P1 PINN optimisation (beam)	Why long-window damped-vibration PINNs fail; window law + reference-free rule	Diagnostics done (E01–E04); window sweep paused
P2 Railway transfer + Physics Reference Layer	Verification-labelled physics references for railway models	Not started
P3 Railway application	Predictive maintenance use case (axle / bearing / wheel)	Not started

No railway claim is made yet: all evidence so far is on the beam benchmark.

Benchmark: a damped fixed-fixed Euler–Bernoulli beam with an exact analytical solution

Quantity	Value
Boundary conditions	fixed-fixed, mode 1
Damped frequency ω_d	129.32 rad/s (20.58 Hz)
Window	1 s = 20.58 cycles
Exact decay rate ζ	3.54 1/s
Reference	analytical modal solution (exact)
Success =	full-window persistence (20.58 cycles) + correct decay + low L2

Batch 1's best configuration (B1) reproduces the frequency but over-damps and collapses at 8.1 of 20.6 cycles

B1 = hard IC/BC + $\tanh^2(\omega_1 t)$ + D4 Fourier + exp-decay LR + mini-batch 128 (Z4-20K)	Value
Training	20,000 steps, 2.56M PDE evaluations, 2,290 s
L2 (exact)	0.263
Fitted frequency	129.2 rad/s
Collapse	0.393 s = 8.1 of 20.6 cycles
Decay	5.40 vs 3.54 1/s (over-damped)
Verdict	FAIL: no validated optimised PINN

HYPOTHESES ELIMINATED

Five candidate explanations were tested at matched compute and eliminated

Hypothesis	Test	Key number	Verdict
Floating-point precision	FP64 control (E01)	identical L2 0.526 / 0.549 / 0.402 in 3 seeds	not material
4th-order operator	mixed form, order 2 (E01)	persistence 1.67–2.13 vs B1 3.11–5.10 cycles	cheaper, not better
Collocation coverage	resampling / 4× reservoir (E03)	H1 supported in 1/3 seeds only; 4X leaves front stalled	insufficient (CASE D)
Optimiser: full-batch Adam	ADAM-FULL (E03)	H2 falsified 0/3	eliminated
L-BFGS on the full field	Adam→L-BFGS transfer (E02)	worse than B1 in 3/3; L2 ×1.28–1.54	eliminated

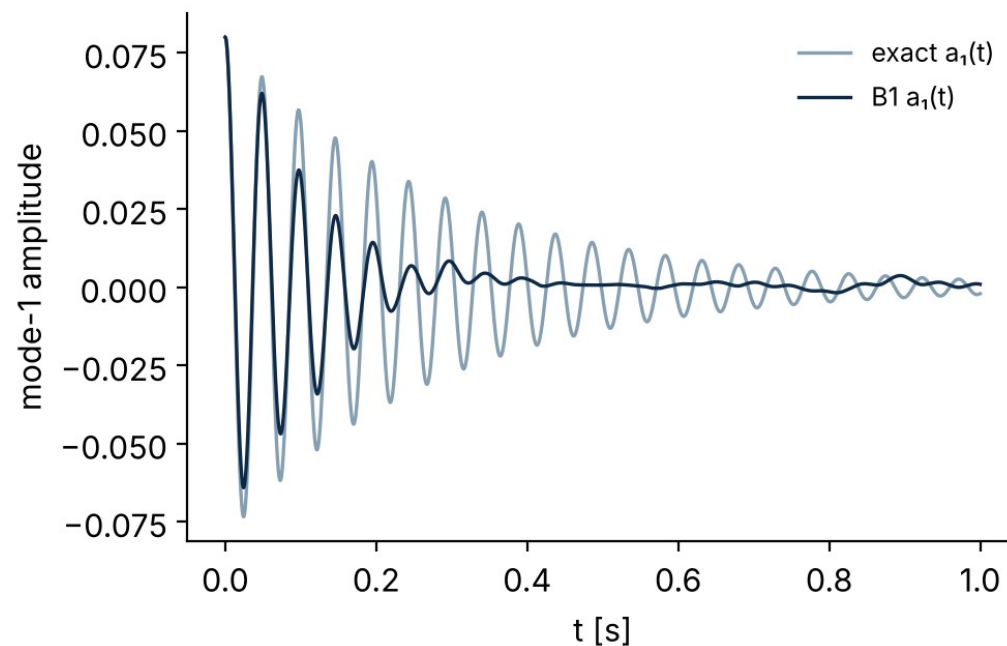
The same optimiser solves the modal problem but fails on the full field

Seed	Modal Adam→L-BFGS: cycles / L2	Full-field B1: cycles / L2	Full-field Adam→L-BFGS: cycles / L2
1234	20.58 / 0.038	3.15 / 0.526	1.60 / 0.671
1235	20.58 / 0.071	3.11 / 0.549	0.72 / 0.843
1236	20.58 / 0.093	5.10 / 0.402	2.53 / 0.578

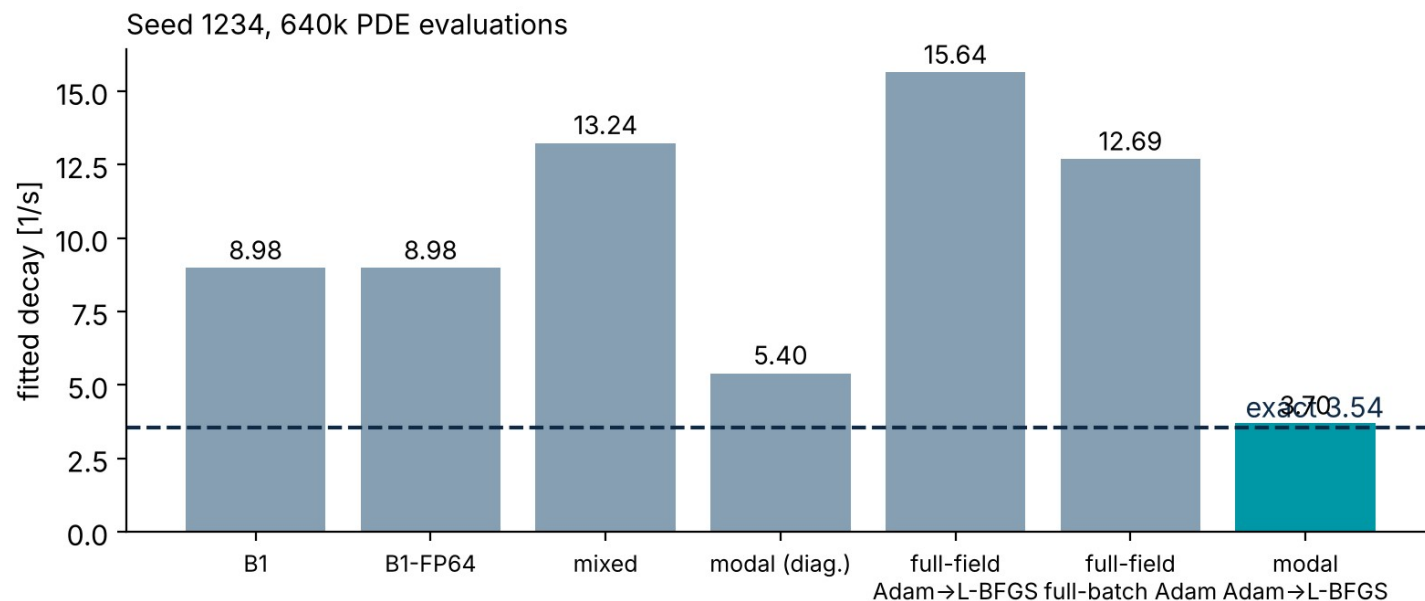
- Modal: full window (20.58 cycles) in 3/3 seeds — but the exact mode shape is supplied (diagnostic, not a method).
- Full field: worse than B1 in 3/3 seeds (persistence −1.6 to −2.6 cycles).
- Decision E: the improvement exists only in the modal diagnostic.

Collapse is a loss of first-mode amplitude; no candidate diagnostic leads it (CASE F)

- Residual energy outside mode 1: $1.1 \cdot 10^{-6}$ (B1, 640k, seed 1234); higher modes $\leq 5 \cdot 10^{-5}$ of exact energy in all runs.
- Collapse = $a_1(t) \rightarrow 0$, not growth of higher modes.
- Spatial/temporal gradient ratio: $1.6\text{--}3.4 \cdot 10^4$ at init, $0.87\text{--}2.0$ after training (no late imbalance).
- Verdict: CASE F — no clear mechanism among A–D (B partial, not discriminating).



Every failing arm decays too fast: artificial damping is the common symptom



Proposed damping-bias law (long windows):

$$\lambda^* = \sqrt{(\zeta^2 + (R / 2\omega)^2)}$$

R = unresolved residual per unit amplitude [s⁻²], measurable without the reference.

Only the modal L-BFGS arm (3.70) is near the exact 3.54.

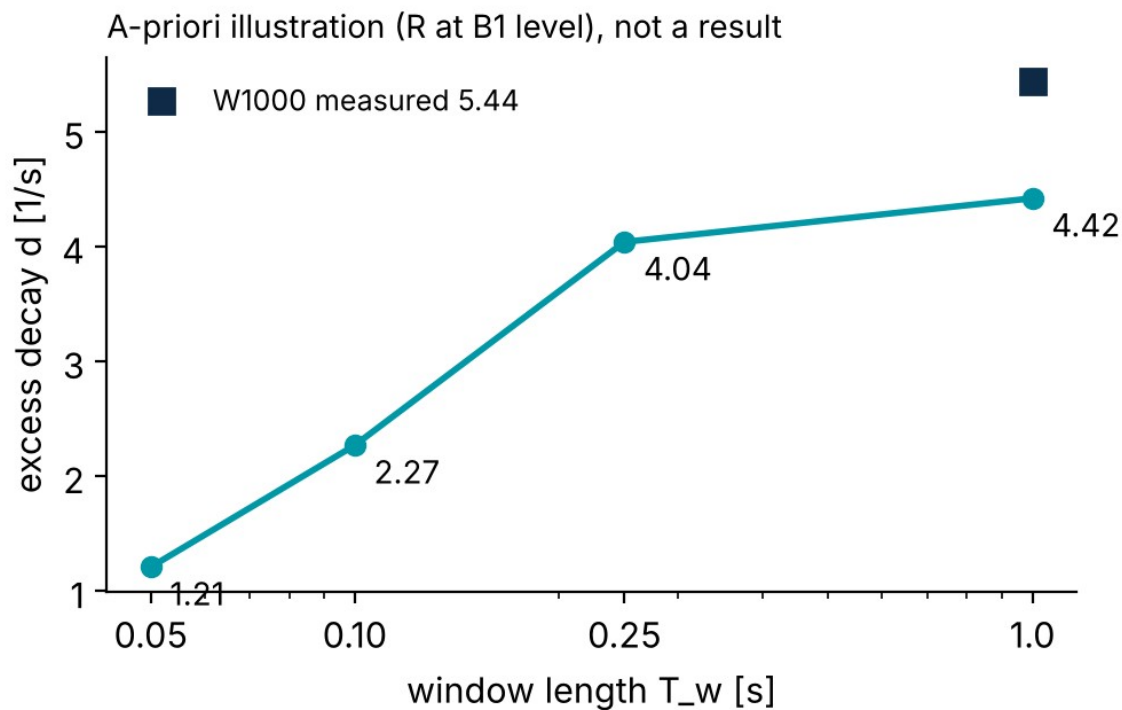
Status: hypothesis under test (B2-E05).

Reweighting the loss moves the damping in both directions, but none beat B1

Variant	L2_exact	Fitted decay (exact 3.54)	Behaviour
B1 (logged)	0.526	8.98	over-damped
normalised loss (degree 0)	5.48 @2500 steps	growing	amplitude ratio 1.8 / 9.3 / 61 at t = 0.05 / 0.4 / 1.0 s
fixed exponent p = 0.5	0.818	10.20	late spurious growth
energy-rate loss $\mu = 10$	4.49 @2500 steps	—	collapsed to a static field
feedback controller on p	0.604	11.19	sensor latched onto growth; p driven to 0

Takeaway: the loss's amplitude homogeneity sets the sign of the bias; the remaining lever with a quantitative prediction is window length.

Window law: shorter windows should cut excess decay; first run W1000 reproduces B1 bit-for-bit



W1000 (1 window, seed 1234): bit-identical to logged B1, L2 0.52639.

Measured excess decay 5.44 1/s vs predicted 6.29 from its measured R (2374 s⁻²).

Prior art: AT-PINN (2024) and AT-PINN-HC (2025) do time-marching with hard constraints on EB beams, but give no relation between segment length and damping error.

Novelty claimed only for the law + reference-free window rule, if validated.

Status: diagnostics are validated; the window law is untested beyond one window

Validated (pre-registered, 3 seeds)	Exploratory (seed 1234)	Pending
B1 reproduced bit-exactly in PINN-RP	loss-homogeneity runs (6 variants)	W050 / W100 / W250 sweep: not yet run (W050 interrupted at 11/20 windows)
FP64, mixed, coverage, L-BFGS transfer eliminated	W1000 vs prediction (1 run)	Block-3 gate: prediction vs measurement
Collapse = mode-1 amplitude loss (E04, CASE F)		Seeds 1235/1236; sweep currently PAUSED awaiting PI

No optimised PINN is claimed.

TIMELINE

Timeline: P1 manuscript by 20 Oct, P2 and P3 by 20 Nov, each behind a gate

Milestone	Date	Gate
Window sweep (W050/W100/W250) + Block-3 analysis	on resume	PASS / PARTIAL / FAIL pre-registered in B2-E05.md
P1 manuscript (beam)	20 Oct	PASS → law + rule; FAIL → diagnostic manuscript
P2 railway transfer + Physics Reference Layer	by 20 Nov	supervisor decisions (next slide) + data access
P3 railway application	by 20 Nov	validated transfer model

PLACEHOLDER

Railway transfer model: axle, bearing and wheel dynamics

to be added

Decisions needed from the supervisor

- Target vehicle
- Sensor scenario: on-board or wayside
- Bearing type and drawing
- Validation data source
- Does a “submitted” paper count for graduation?